C4.1 Populations and communities

NAME	DATE	TOTAL
		_____ / 50

About 45 minutes, or one section at a time. The number of marks shows how many separate points to make. You may use a calculator.

Section A: Recall

12 MARKS

1 DEFINE population. [C4.1.1] [2]

2 DEFINE carrying capacity. [C4.1.5] [1]

3 STATE two density-dependent factors. [C4.1.6] [2]

4 LIST the three phases of a sigmoid population growth curve, in order. [C4.1.7] [3]

5 STATE one named example of parasitism and one of pathogenicity. [C4.1.11] [2]

6 OUTLINE one named example of allelopathy. [C4.1.18] [2]

Section B: Understanding

18 MARKS

7 EXPLAIN why quadrats must be placed randomly. [C4.1.2] [2]

8 OUTLINE how random quadrats are used to estimate the population of a sessile organism. [C4.1.3] [3]

9 STATE two assumptions made when using the Lincoln index. [C4.1.4] [2]

10 **DISTINGUISH** between intraspecific competition and cooperation, using an example of each. [C4.1.10] [3]

11 **EXPLAIN** how a hard coral and its zooxanthellae each benefit from their relationship. [C4.1.12] [2]

12 **EXPLAIN** how an invasive species can cause an endemic species to decline. [C4.1.13] [2]

13 **DESCRIBE** one field method of testing for interspecific competition. [C4.1.14] [2]

14 **DISTINGUISH** between top-down control and bottom-up control. [C4.1.17] [2]

Section C: Working with data

14 MARKS

All data in this section are invented for practice. The patterns are realistic.

15 [C4.1.3]

Limpets were counted in eight random 0.25 m^2 quadrats on a rocky shore with an area of 50 m^2.

QUADRAT	1	2	3	4	5	6	7	8
LIMPETS	4	7	2	5	6	3	8	5

a **CALCULATE** the population of limpets on the shore. Show your working. [2]

b **OUTLINE** what the standard deviation of these counts would show. [1]

16 [C4.1.4]

60 ground beetles were caught in pitfall traps, marked with a dot of paint and released. Two days later, 45 beetles were caught, and 9 of them were marked.

a **CALCULATE** the population size. [1]

b **EXPLAIN** the effect on the estimate if some paint marks wore off before the second catch. [2]

17 [C4.1.15]

The presence or absence of two meadow plants, X and Y, was recorded in 60 random quadrats.

OBSERVED	Y PRESENT	Y ABSENT	TOTAL
X PRESENT	6	24	30
X ABSENT	18	12	30
TOTAL	24	36	60

a **STATE** the null hypothesis. [1]

b **CALCULATE** the four expected values. [2]

c **CALCULATE** the value of chi-squared. Show your working. [2]

d **DEDUCE** what the result shows. The critical value for 1 degree of freedom at p = 0.05 is 3.84. [2]

e **SUGGEST** one reason for the result. [1]

Section D: Longer answer

6 MARKS

18 **EXPLAIN** how density-dependent factors control the size of a population, using a named predator–prey example. [C4.1.6, C4.1.7, C4.1.16] [6]